

# Math 104 – Lecture 02

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September 7, 2026

## 1 Doing $\epsilon - N$ proofs

**Example 1.1** (1). Show that  $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$ .

*Proof.* Square it, starting with

$$\left| \frac{1}{\sqrt{n}} - 0 \right| < \epsilon$$
$$\frac{1}{n} < \epsilon^2 \rightarrow n > \frac{1}{\epsilon^2}$$

□

You define  $N = \frac{1}{\epsilon^2}$ , and when  $n > N$  the limit holds.

**Example 1.2** (2). Show that  $\lim_{n \rightarrow \infty} \frac{3n+1}{7n+4} = \frac{3}{7}$

*Proof.* We do this by writing the format first

$$\left| \frac{3n+1}{7n+4} - \frac{3}{7} \right| < \epsilon$$

□

We take a common denominator, to get

$$\left| \frac{(3n+1)7 - 3(7n+4)}{(7n+4)(7)} \right| < \epsilon$$

Then, this simplifies to  $\left| \frac{19}{(7n+4)(7)} \right|$  which we can pull out absolute value as we see  $(7n+4)$  is positive. The inequality then simplifies to  $\frac{19}{49\epsilon} + \frac{4}{7} < n$  and we set  $N = \frac{19}{49\epsilon} + \frac{4}{7}$

**Proof Format:**

To do  $\epsilon - N$  proofs, you just define  $N =$  something in terms of  $\epsilon$ . Then, set  $n > N$ , and show that the absolute value expression can then be rearranged to show the original expression.

**Example 1.3 (3).** Show that  $\lim_{n \rightarrow \infty} \frac{4n^2+3n}{n^2-6} = 4$

**Scratch Work:** We want the inequality

$$\left| \frac{4n^2 + 3n}{n^2 - 6} - 4 \right| < \epsilon$$

We take the common denominator:

$$\left| \frac{(4n^2 + 3n) - 4(n^2 - 6)}{n^2 - 6} \right|$$

which is

$$\left| \frac{3n^2 + 24}{n^2 - 6} \right| < \epsilon \rightarrow \text{notice that both numerator and denominator go to infinity}$$

So we can remove the absolute value. Something here we do is we notice  $n > 1$  so we write

$$\frac{3n + 24}{n^2 - 6} < \frac{27n}{n^2 - 6} < \frac{27n}{\frac{1}{2}n^2} < \frac{54}{n}$$

for sufficiently large  $n$ .

So you just do  $N = \max\{\frac{54}{\epsilon}, 12\}$ .

## 2 Definitions and notation

**Definition 2.1** (First definition). Write the definition and one sentence explaining why it matters.

## 3 Claims, theorems, and proof sketches

**Claim.** State the claim in one precise sentence.

**Proof sketch.** Record the key idea, the first useful equation, and the step that still needs checking.

## 4 Examples and calculations

**Example 4.1** (Lecture example). Write the setup, the calculation, and the conclusion.

## 5 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: